Threat Hunt Report

Title: Internal Network Reconnaissance
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Platform: Microsoft Sentinel / Microsoft Defender
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To prevent exposure of sensitive internal telemetry, specific device names, IP addresses, and raw query outputs have been omitted from this report. However, the summarized findings and behavioral patterns observed during the investigation are documented below.

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1. Objective

The objective of this threat hunt was to identify potential **internal reconnaissance activity** within the environment.

Adversaries who gain initial access often perform network discovery to identify reachable systems and services before attempting lateral movement.

This hunt aimed to detect behaviors associated with:

- internal host discovery
- port scanning
- abnormal connection patterns

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2. Threat Hypothesis

If an attacker has compromised a host within the environment, the adversary may perform **internal network reconnaissance** by scanning multiple systems or services.

This activity may generate unusual network behavior such as:

- connections to many unique IP addresses
- connections across many ports

- high volumes of connection attempts
- rapid connection bursts within short time windows

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3. MITRE ATT&CK Mapping

Technique: **T1046 – Network Service Discovery**

Description:

Adversaries may attempt to discover network services running on internal systems in order to identify targets for lateral movement.

This activity typically involves scanning hosts for open ports or reachable services.

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4. Data Sources

The following telemetry sources were used during the hunt:

Data Source	Description
DeviceNetworkEvents	Endpoint network connection telemetry
Microsoft Defender for Endpoint	Process and network monitoring
Microsoft Sentinel	Log analytics and threat hunting

Primary table used:

DeviceNetworkEvents

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5. Hunting Methodology

The hunt was conducted using a progressive investigation approach.

The analysis focused on identifying:

1. Hosts communicating with large numbers of unique IP addresses
2. Hosts communicating across many destination ports
3. Hosts generating unusually high connection volumes
4. Processes responsible for network activity
5. Destination port distribution patterns

Each step was validated using additional queries and baseline comparison.

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6. Query 1 — Unique IP Analysis

The first query identified hosts communicating with large numbers of unique IP addresses.

```
DeviceNetworkEvents

| where Timestamp > ago(24h)

| summarize UniqueIPs = dcount(RemoteIP) by DeviceName

| where UniqueIPs > 100

| sort by UniqueIPs desc
```

Observation

Several endpoints communicated with more than **2000 unique IP addresses** within the observation period.

Further classification revealed that the majority of these connections were directed toward **public IP addresses**, indicating normal internet browsing and cloud service traffic.

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7. Query 2 — Port Diversity Analysis

To identify potential port scanning activity, hosts communicating with many ports were analyzed.

```
DeviceNetworkEvents

| where Timestamp > ago(24h)

| summarize UniquePorts = dcount(RemotePort) by DeviceName, RemoteIP

| where UniquePorts > 50

| sort by UniquePorts desc
```

Observation

Results showed several **virtual machines** communicating across many ports.

Further investigation revealed that the activity originated from:

- system processes
- infrastructure communication
- application services

No evidence of malicious port scanning behavior was identified.

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8. Query 3 — Connection Volume Baseline

The third query analyzed hosts generating large numbers of connections.

```
DeviceNetworkEvents

| where TimeGenerated > ago(24h)

| summarize

  connectioncount=count(),

  uniqueip=dcount(RemoteIP),

  uniqueports=dcount(RemotePort) by DeviceName

| where connectioncount > 500

| sort by connectioncount desc
```

Observation

Endpoints generated between **5500 and 7600 network connections within 24 hours.**

Typical endpoint behavior included:

- 1500–2100 unique IP addresses
- 17–40 unique ports

This pattern is consistent with normal workstation activity driven by:

- web browsing
- cloud services
- operating system telemetry

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9. Process Attribution Analysis

Processes responsible for network activity were examined.

Common processes included:

- `chrome.exe`
- `msedge.exe`
- `firefox.exe`
- `svchost.exe`
- `onedrive.exe`
- `mssense.exe`

These processes correspond to normal endpoint behavior such as:

- web browsing
- system services
- cloud synchronization
- endpoint security telemetry

A review of PowerShell activity revealed **only a single network connection**, indicating no abnormal scripting behavior.

10. Destination Port Distribution Analysis

Destination ports were analyzed to identify potential scanning patterns.

Results showed the following distribution:

Port	Service
443	HTTPS
53	DNS
80	HTTP
5357	Windows service discovery
5358	Windows service discovery
7680	Windows Delivery Optimization
5228	Google service communication

The majority of traffic was directed toward **standard web and network service ports.**

This pattern is consistent with normal endpoint internet usage.

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11. Network Destination Investigation

Further analysis identified several high-volume connection destinations including:

- **20.x.x.x:445 –** Microsoft Azure cloud infrastructure
- **10.x.x.x:53 –** internal DNS server
- **169.254.169.254 –** cloud instance metadata service

These connections were consistent with:

- cloud storage synchronization
- DNS resolution
- cloud virtual machine metadata access

No malicious destinations were identified.

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12. Findings

The investigation revealed that network activity within the environment was consistent with normal endpoint operations.

Observed behavior included:

- web browsing traffic
- cloud application connectivity
- DNS resolution
- operating system service discovery

No indicators of internal reconnaissance or port scanning were detected.

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13. Conclusion

The threat hunt did not identify evidence of malicious internal reconnaissance activity.

All observed network patterns were consistent with normal enterprise endpoint behavior.

The hunt provided valuable insights into baseline network behavior within the environment.

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14. Recommendations

The following improvements are recommended to enhance detection capabilities:

- Implement detection rules for rapid internal connection bursts
- Monitor hosts communicating with unusually high numbers of internal IP addresses
- Track SMB and RDP connections between internal systems
- Continue periodic threat hunting activities to identify anomalous behavior

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Recommendations for Future hunts

Future hunts should incorporate time-based burst analysis and lateral movement monitoring to improve detection of rapid reconnaissance activity.